PAUL M. CHICHURA

Curriculum Vitae (May 2026)

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APPOINTMENTS

2025-*current* **Postdoctoral Scholar**, NSF-Simons AI Institute for the Sky (SkAI), University of Chicago
Supervisor: Alex Drlica-Wagner

EDUCATION

2025 **Ph.D., Physics**, *(Nathan Sugarman Award, Sachs Fellowhip)*, University of Chicago
Advisor: Thomas Crawford

2022 **M.S., Physics**, University of Chicago

2018 **M.S., Physics and Astronomy**, University of Pennsylvania

2018 **B.A., Physics**, *(Summa Cum Laude, dept. honors, Phi Beta Kappa)*, University of Pennsylvania
Advisor: James Aguirre

2014–2015 Schreyer Honors College, The Pennsylvania State University

RESEARCH EXPERIENCE

2025–*current* **NSF-Simons AI Institute for the Sky (SkAI)**
- Developing, AI algorithms for **scheduling astronomical surveys**

2020–2025 **South Pole Telescope (SPT) and Event Horizon Telescope (EHT)**
- Created, deployed ML models to improve **pointing accuracy** of SPT for EHT observations
- Designed new surveys of the **galactic center** with SPT
- Led analysis of monitoring campaign of **variability of Sgr A*** with SPT
- Developed tools for the first targeted analysis of **asteroids** in data from CMB surveys

2017–2019 **Hydrogen Epoch of Reionization Array (HERA)**
- Wrote pipeline to make first power spectra for data from the HERA commissioning array

HONORS AND AWARDS

2024 Nathan Sugarman Award for Excellence in Graduate Student Research, University of Chicago

2021 Sachs Fellowship, University of Chicago

2021 Chambliss Astronomy Achievement Student Award Honorable Mention, American Astronomical Society

2018 Phi Beta Kappa Honor Society, University of Pennsylvania

PROFESSIONAL SERVICE AND LEADERSHIP

2026-*current* Co-chair, Local Organizing Committee, Open SkAI 2026 Conference, SkAI

2024-2025 Analysis Coordinator, Sources and Transients Working Group, SPT Collaboration, University of Chicago

2022-2025 Program Lead, First Discoveries outreach initiative, SPT Collaboration, University of Chicago

2021-2024 Coordinator, Climate Surveys, Chicago SPT Collaboration, University of Chicago

RESEARCH MENTORSHIP

Graduate Students (1):

2023-*current* Bhaskar Mondal *(project advisor, non-voting committee member)* – Ph.D. student in Aerospace Engineering at the University of Illinois Urbana-Champaign. Detecting and thermal modeling asteroids in SPT CMB data.

Undergraduate Students (3):

2023 Mariah Jones *(co-supervised with Thomas Crawford)* – Astrophysics major at Vassar College, summer REU at University of Chicago. Thermal modeling of asteroids with microwave data.

2022 Marvin Lopez Acevedo *(co-supervised with Alexandra Rahlin)* – Physics and Geosciences major at Hamilton College, summer REU at University of Chicago. Building monitor webpages for SPT-SZ cryostat.

2021 Nahuel Ossa-Jaen *(co-supervised with Thomas Crawford)* – Physics and Computer Science major at University of Redlands, summer REU at University of Chicago. Detecting asteroids in SPT CMB data.

TEACHING

2025 College Teaching Certificate, University of Chicago
2024 Guest Lecturer, University of Chicago, *ASTR 12620: The Big Bang*
2019–2020 Teaching Assistant, University of Chicago
PHSC 116: Physics for Future Presidents, *PHSC 117: Physics for Future Presidents*, *PHYS 123: General Physics III*
2018–2019 Teaching Assistant, University of Pennsylvania
PHYS 101: General Physics: Mechanics, *PHYS 102: General Physics Electricity and Magnetism*

COMMUNITY OUTREACH AND PUBLIC EDUCATION

2020–2025 Program Lead & Classroom Volunteer, First Discoveries**,** University of Chicago
- SPT collaboration initiative to improve early-childhood science education and teacher self-efficacy
- led volunteer meetings, communicated with school administration, published select lesson plans[1]
- designed and taught lessons at John Fiske Elementary School, primary contact for four classrooms
- co-designed, co-led 2 professional development trainings for 15 teachers each, 2023 and 2024

2022 Volunteer, South Side Science Festival, UChicago
2021–2023 Discussion Organizer, Chicago SPT Collaboration, University of Chicago
- Organized and led weekly discussions on inclusivity efforts during local lab group meetings
- Initiated, conducted 3 climate surveys of the Chicago-SPT group

CONFERENCE PRESENTATIONS

2026 **Invited Speaker,** CosmicAI's Cosmic Horizons 2026, "Self-Driving Telescopes: Intelligent Scheduling for Astronomical Surveys," National Radio Astronomy Observatory, 14-16 Jul. 2026.
2026 **Speaker,** National Radio Science Meeting, "Telescope Control with Machine Learning: Pointing Corrections and Observation Scheduling," University of Colorado Boulder, 6-9 Jan. 2026.
2025 **Speaker,** National Radio Science Meeting, "Pointing the South Pole Telescope with Machine Learning," University of Colorado Boulder, 7-10 Jan. 2025.
2024 **Poster,** AI+Science, "Pointing Model Predictions with Machine Learning for the South Pole Telescope," University of Chicago, 15-19 Jul. 2024.
2023 **Poster,** AI+Science, "Pointing Model Predictions for the South Pole Telescope," UChicago, 17-21 Jul. 2023.
2023 **Speaker,** The Transient and Variable Universe 2023, "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope," University of Illinois Urbana-Champaign, 20-22 Jun. 2023.
2021 **Poster**, 238th Meeting of the American Astronomical Society, "Measuring the Millimeter-Wavelength Flux of Asteroids with the South Pole Telescope," remote, 7-9 Jun 2021.
2018 **Poster**, 231st Meeting of the American Astronomical Society, "Polarized Power Spectra from HERA-19 Commissioning Data: Effect of Calibration Techniques," Washington, DC, 8-9 Jan. 2018.

SELECTED PUBLICATIONS

[1] Wan, Y., J. Vieira, **P. Chichura**, *et al.* 2026. "Detection of Millimeter-Wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey." *ApJ* 997, no. 2 (February): 133. arXiv:2509.08962.
- Designed and led the observation survey that resulted in data analyzed in this paper

[2] **Chichura, P.**, *et al.* 2025. "Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning." *Journal of Astronomical Instrumentation* 14 (January): 2550001. arXiv:2412.15167.
- Led analysis, development, and deployment of models; a substantial part of Ph.D. thesis

[3] **Chichura, P.**, *et al.* 2022. "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope." *ApJ* 936, no. 2 (September): 173. arXiv:2202.01406.
- Led analysis and development of novel techniques; a substantial part of Ph.D. thesis

[4] Kohn, S., *et al.* 2019. "The HERA-19 Commissioning Array: Direction-dependent Effects." *ApJ* 882, no. 1 (September): 58. arXiv:1802.04151.
- Results from undergraduate honors thesis research

[1] https://pole.uchicago.edu/public/First%20Discoveries.html

ALL PUBLICATIONS

[5] Archipley, M., *et al.* 2026. "Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs." *A&A* 706 (January): A17. arXiv:2506.00298.

[6] Bernshteyn, V., *et al.* 2026. "Ring Asymmetry and Spin in M87*." *ApJ* 1000, no. 2 (April): 231. arXiv:2601.00394.

[7] Camphuis, E., *et al.* 2026. "SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field." *PhRvD* 113, no. 8 (April): 083504. arXiv:2506.20707.

[8] Chaubal, P., *et al.* 2026. "SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power." *arXiv e-prints* (January): arXiv:2601.20551.

[9] de Haan, T., *et al.* 2026. "Characterization of the Polarization Beam Response of SPT-3G Using Point Sources." *arXiv e-prints* (February): arXiv:2602.06334.

[10] Georgiev, B., *et al.* 2026. "Locating the missing large-scale emission in the jet of M87* with short EHT baselines." *A&A* 709 (April): A26. arXiv:2601.13356.

[11] Maniyar, A. S., *et al.* 2026. "SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys." *arXiv e-prints* (February): arXiv:2602.11279.

[12] Qu, F., et al. 2026. "Unified and consistent structure growth measurements from joint ACT, SPT and Planck CMB lensing." *PhRvL* 136, no. 2 (January): 021001. arXiv:2504.20038.

[13] Quan, W., *et al.* 2026. "SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field." *arXiv e-prints* (March): arXiv:2603.20163.

[14] Raghunathan, S., *et al.* 2026. "Measurement of the Full Shape of the Thermal SunyaevZel'dovich Power Spectrum from the South Pole Telescope and HerschelSPIRE Observations." *ApJL* 1002, no. 1 (May): L22. arXiv:2602.10107.

[15] Saurabh, H. *et al.* 2026. "Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations." *A&A* 706 (January): A27. arXiv:2512.08970.

[16] Tandoi, C., *et al.* 2026. "Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data." *arXiv e-prints* (May): arXiv:2605.01022.

[1] Wan, Y., J. Vieira, **P. Chichura**, *et al.* 2026. "Detection of Millimeter-Wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey." *ApJ* 997, no. 2 (February): 133. arXiv:2509.08962.

[2] **Chichura, P.**, *et al.* 2025. "Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning." *Journal of Astronomical Instrumentation* 14 (January): 2550001. arXiv:2412.15167.

[17] Coerver, A., *et al.* 2025. "Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G." *ApJ* 982, no. 1 (March): 15. arXiv:2407.20579.

[18] Event Horizon Telescope Collaboration. 2025. "Horizon-scale variability of M87* from 2017–2021 EHT observations." *A&A* 704 (December): A91. arXiv:2509.24593.

[19] Foster, A., *et al.* 2025. "Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites." *The Open Journal of Astrophysics* 8 (May): 51. arXiv:2411.03374.

[20] Ge, F., *et al.* 2025. "Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data." *PhRvD* 111, no. 8 (April): 083534. arXiv:2411.06000.

[21] Khalife, A., *et al.* 2025. "SPT-3G D1: Axion Early Dark Energy with CMB experiments and DESI." *arXiv e-prints* (July): arXiv:2507.23355.

[22] Kornoelje, K., *et al.* 2025. "The SPT-Deep Cluster Catalog: Sunyaev-Zel'dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees." *arXiv e-prints* (March): arXiv:2503.17271.

[23] Vitrier, A., *et al.* 2025. "Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky." *arXiv e-prints* (October): arXiv:2510.24669.

[24] Zebrowski, J., *et al.* 2025. "Constraints on Inflationary Gravitational Waves with Two Years of SPT-3G Data." *PhRvD* 112, no. 12 (December): 123520. arXiv:2505.02827.

[25] Ansarinejad, B., *et al.* 2024. "Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing." *JCAP* 2024, no. 7 (July): 024. arXiv:2404.02153.

[26] Prabhu, K., *et al.* 2024. "Testing the ΛCDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G." *ApJ* 973, no. 1 (September): 4. arXiv:2403.17925.

[27] Raghunathan, S., *et al.* 2024. "First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel'dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations." *PhRvL* 133, no. 12 (September): 121004. arXiv:2403.02337.

[28] Tandoi, C., *et al.* 2024. "Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G." *ApJ* 972, no. 1 (September): 6. arXiv:2401.13525.

[29] Balkenhol, L., *et al.* 2023. "Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset." *PhRvD* 108, no. 2 (July): 023510. arXiv:2212.05642.

[30] Pan, Z., *et al.* 2023. "Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data." PhRvD 108, no. 12 (December): 122005. arXiv:2308.11608.

[31] Schiappucci, E., *et al.* 2023. "Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel'dovich effect with SPT-3G and DES." *PhRvD* 107, no. 4 (February): 042004. arXiv:2207.11937.

[3] **Chichura, P.**, *et al.* 2022. "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope." *ApJ* 936, no. 2 (September): 173. arXiv:2202.01406.
[32] Ferguson, K., *et al.* 2022. "Searching for axionlike time-dependent cosmic birefringence with data from SPT-3G." *PhRvD* 106, no. 4 (August): 042011. arXiv:2203.16567.
[23] Ghosh, A., *et al.* 2020. "Foreground modelling via Gaussian process regression: an application to HERA data." *MNRAS* 495, no. 3 (January): 2813–2826. arXiv:2004.06041.
[4] Kohn, S., *et al.* 2019. "The HERA-19 Commissioning Array: Direction-dependent Effects." *ApJ* 882, no. 1 (September): 58. arXiv:1802.04151.